A* Examination Guide

Built from 16 past papers and their mark schemes, June 2019 to June 2026

This guide is built from the Paper 1H question papers and mark schemes. Methods are laid out the way the mark schemes award M, A and B marks, and every warning comes from a mark-scheme note such as "working required", "do not ISW", "SC" or "correct answer from obvious incorrect working scores 0". The formulae sheet is printed in the paper; everything else must be learnt.

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Papers analysed: Jun 2019, Jan 2020, Nov 2020, Jan 2021, Jun 2021, Nov 2021, Jan 2022, Jun 2022, Jan 2023, Jun 2023, Nov 2023, Jun 2024, Nov 2024, Jun 2025, Nov 2025, Jun 2026. Papers are cited by series, e.g. (Jun 25).

1 HOW PAPER 1H IS BUILT

Format: 2 hours, 100 marks, 20–26 questions, calculator allowed. About 1.2 minutes per mark. Questions rise in difficulty: the first ten are grade 4–6 skills (mean from a table, standard form, percentages, factorising); the last eight are grade 8–9 (vectors, functions, non-linear simultaneous equations, bounds, calculus, 3D trigonometry). "You must write down all the stages in your working."

Topic	Papers (of 16)	What is asked most
Indices and algebraic manipulation	16	simplify powers, expand triple brackets, factorise, rearrange
Percentages (reverse, compound)	16	original price, multi-year change, find the rate x%
Probability (trees, without replacement)	16	complete a tree, "at least one", three items taken
Trigonometry, sine and cosine rules	15	two-triangle problems, area ½ab sin C, bearings
Arithmetic sequences and series	14	nth term, S_n, two simultaneous conditions
Standard form	14	convert, calculate, give the answer in standard form
Bounds	12	upper/lower bound of a formula; "suitable degree of accuracy"
Histograms and cumulative frequency	12	frequency density, estimate from bars, median, IQR
Vectors	12	express a vector; use a vector method to find a ratio or n
Proportion	12	$y \propto x^2$, $y \propto 1/x^3$: find k, then use the formula
Sectors and circles	12	arc length, sector area, segment area or perimeter
Surds	10	simplify √675; rationalise (a + √b) denominators
Functions	10	f(a), inverse, composite, excluded value, solve fg(x) < 0

Topic	Papers (of 16)	What is asked most
Calculus	9	gradient = k, turning points, maximum volume, kinematics, tangent at a point
Non-linear simultaneous equations	8	line and quadratic or circle
Graph transformations	8	turning point of f(x + a), f(x) + b, f(ax), −f(x); trig graphs

2 TWENTY MARK-SCHEME RULES

1. **"Working required":** for these questions a correct answer with no method scores **0** (e.g. polygon angles, trapezium area, rationalising, fg(x) < 0, simultaneous equations in Jun 25).
2. **No working, wrong answer = 0.** "If no working is shown then incorrect (even though nearly correct) answers score no marks." Write every step.
3. **Correct answer from obviously incorrect working scores 0.**
4. **Do not spoil a correct answer:** "do not ISW incorrect simplification" – (17x − 6)/12x = 11/12 lost the A mark.
5. **Values in quotation marks** in the mark scheme must come from a correct method seen; show where numbers come from.
6. **Multipliers:** 1 − 0.04 or 0.96 is accepted; "(1 − 4%)" is not (Jun 25).
7. **Rationalising:** multiply numerator and denominator explicitly by the conjugate; all four terms of the denominator expansion must be correct.
8. **Proportion:** write $y = k/x^3$ with a constant; "$y = 1/x^3$" scores nothing for the method.
9. **Trial and improvement scores no marks** unless fully correct, where algebra is required (Jun 21).
10. **Sets of values:** inequalities need the correct sign; "x > 9/2" not "x = 9/2".
11. **Bounds:** show the bounds used; the upper bound of a − b uses upper a and lower b; for a/b use upper a and lower b. "Suitable degree of accuracy" needs a reason (both bounds round to the same value).
12. **Graphs:** plot points accurately (within the overlay circles); draw a smooth curve, not straight segments.
13. **Histograms:** frequency = frequency density × class width; label the axis frequency density.
14. **Probability trees:** all branch pairs must add to 1; answers as fractions, decimals or percentages, never ratios.
15. **Accuracy:** give the d.p. or s.f. asked; keep at least 4 s.f. in working; π from the calculator (3.14 or 22/7 often accepted for the method only).
16. **Angles with reasons:** circle-theorem questions need each reason stated in full ("angle in a semicircle is 90°").
17. **Vector methods:** form two expressions for the same vector, equate coefficients of a and b, and solve. A guessed ratio without a vector method scores at most the first mark.
18. **Quadratic solutions:** when a quadratic is solved by formula, show the substitution; a calculator answer from an incorrect quadratic scores no accuracy marks.
19. **Standard form:** $a \times 10^n$ with $1 \le a < 10$; 35×10^{-138} is not standard form.
20. **Answer on the answer line;** if the working gives a better answer, the answer line is still marked.

3 NUMBER: POWERS, STANDARD FORM, SURDS, BOUNDS

INDEX LAWS

$a^m \times a^n = a^{m+n}$; $a^m \div a^n = a^{m-n}$; $(a^m)^n = a^{mn}$; $a^0 = 1$; $a^{-n} = 1/a^n$; $a^{1/n} = \sqrt[n]{a}$; $a^{m/n} = (\sqrt[n]{a})^m$.

Jun 25: $(5a^4r^2)^3 = 125a^{12}r^6$. **Jun 21:** $2 \times 16 = 2^x \Rightarrow x = 5$; write each number as a power of the same prime first.

SURDS

$\sqrt{675} = \sqrt{(225 \times 3)} = 15\sqrt{3}$ (Nov 24). Rationalise with the conjugate: $21/(3 - \sqrt{2}) \times (3 + \sqrt{2})/(3 + \sqrt{2}) = 21(3 + \sqrt{2})/7 = 9 + 3\sqrt{2} = 9 + \sqrt{18}$ (Jun 25).

"Without a calculator" or "show that": every stage must be visible.

BOUNDS

A value correct to the nearest 10 g has bounds ±5 g; to 1 d.p. ±0.05; to 2 s.f. half of the last significant unit.

Jan 21: $a = (v - u)/t$, $v = 9.6$, $u = 3.8$ (1 d.p.), $t = 1.84$ (2 d.p.). Upper bound = $(9.65 - 3.75)/1.835 = 3.22$.

Suitable accuracy (Jan 23, Jun 26): work out both bounds and give the value both round to, with that reason.

- Standard form: 520 million = 5.2×10^8; $(5 \times 10^{42}) \times (7 \times 10^{-180}) = 3.5 \times 10^{-137}$ (Jun 25).
- Prime factors, HCF, LCM: use the product of prime powers; HCF takes the lowest powers, LCM the highest.
- Recurring decimals (Jun 19): $x = 0.25454\ldots$; $1000x - 10x = 252 \Rightarrow x = 252/990 = 14/55$. Show both multiplied lines.

4 PERCENTAGES, RATIO AND PROPORTION

PERCENTAGES

Reverse percentage (Jun 25): price after a 28% reduction is 198 ⇒ original = $198 \div 0.72 = 275$.

Repeated change (Jun 25): $750\,000 \times 0.96 \times 0.935 = 673\,200$; $698\,445 \div 673\,200 = 1.0375 \Rightarrow x = 3.75$.

Compound interest for an unknown rate (Jan 20, Jan 22): divide out the known years, then take the root of the remaining multiplier.

Percentage profit = profit ÷ cost × 100.

DIRECT AND INVERSE PROPORTION

Write the formula with k, substitute to find k, then use it. **Jun 25:** $y \propto 1/x^3$, $y = 4$ when $x = 3 \Rightarrow k = 108$; $y = 864 \Rightarrow x^3 = 1/8 \Rightarrow x = 0.5$.

$y \propto x^2$ doubles x ⇒ y × 4; $y \propto 1/x^2$ doubles x ⇒ y ÷ 4.

- Ratio: total parts, one part, then scale. Combining ratios (Jun 21): make the shared quantity equal (7 : 2 and 5 : 9 → 35 : 10 : 18).
- Speed, density, pressure: $s = d/t$, $\rho = m/V$, $p = F/A$; convert units first (km/h and seconds, cm^3 and m^3).

5 ALGEBRAIC MANIPULATION

- Expand triple brackets in two stages: $5x(x + 2)(3x - 4) = 15x^3 + 10x^2 - 40x$ (Jan 22).
- Factorise fully: take out the highest common factor of numbers and letters; difference of two squares $50g^2 - 18 = 2(5g - 3)(5g + 3)$; quadratics $ax^2 + bx + c$ by splitting the middle term.
- Algebraic fractions: factorise numerator and denominator, then cancel; add with a common denominator: $5/4 + (x - 3)/(6x) = (30x + 4x - 12)/24x = (17x - 6)/12x$ (Jun 25).

- Simplify fully: $(4x^2 - 9)/(2x^2 - 5x - 12) = (2x - 3)(2x + 3)/((2x + 3)(x - 4)) = (2x - 3)/(x - 4)$ (Jun 26).
- Change the subject when it appears twice: collect the subject terms on one side, factorise it out, divide.
- Completing the square: $7 + 12x - 3x^2 = 19 - 3(x - 2)^2$; $5x^2 - 20x + 23 = 5(x - 2)^2 + 3$. The turning point is (2, 19) or (2, 3).

6 EQUATIONS AND INEQUALITIES

NON-LINEAR SIMULTANEOUS EQUATIONS (5 MARKS) · JUN 25

$x^2 + 3y + y^2 = 7$ and $y = x + 2$: substitute $\Rightarrow x^2 + 3(x + 2) + (x + 2)^2 = 7 \Rightarrow 2x^2 + 7x + 3 = 0 \Rightarrow (2x + 1)(x + 3) = 0 \Rightarrow x = -½$, $y = 3/2$ or $x = -3$, $y = -1$. Pair the values correctly.

- Linear equations with fractions: multiply every term by the common denominator first.
- Linear simultaneous equations: eliminate by matching coefficients; substitute back; check in the other equation.
- Quadratic inequality (Nov 25): $2x^2 + x - 28 > 0 \Rightarrow (2x - 7)(x + 4) > 0 \Rightarrow x < -4$ or $x > 3.5$. Sketch the parabola to pick the regions.
- Integer solutions of $-4 \le 2y < 6$: $y = -2, -1, 0, 1, 2$. Regions on a graph: dashed lines for strict inequalities, state each inequality.
- Forming equations from shapes or contexts (Nov 21 tiles, Jun 24 savings): define the variable, write the equation, solve.

7 SEQUENCES AND FUNCTIONS

ARITHMETIC SERIES (ON THE FORMULAE SHEET: $S_N = N/2[2A + (N - 1)D]$)

nth term = $a + (n - 1)d$. **Jun 25:** $a + 19d = 123$ and $15(2a + 29d) = 2880 \Rightarrow 2a + 29d = 192 \Rightarrow d = 6$, $a = 9$.

Terms from the 50th to the 100th: $S_{100} - S_{49}$. Number of terms in a list ending at x: solve $a + (n - 1)d = x$.

Interior angles in an arithmetic sequence (Jun 24): set $S_n = 180(n - 2)$.

FUNCTIONS

fg(x) means apply g first. **Jun 25:** $f(x) = 9 - \sqrt{x}$, $g(x) = 4x^2$: $fg(x) = 9 - 2x < 0 \Rightarrow x > 4.5$.

Inverse: $y = f(x)$, swap, make y the subject: $f(x) = 3x/(2 - x) \Rightarrow f^{-1}(x) = 2x/(3 + x)$.

Excluded value: the x that makes a denominator zero, or a square root negative.

8 COORDINATE GEOMETRY AND GRAPHS

- Gradient = $\Delta y/\Delta x$; midpoint; distance by Pythagoras; perpendicular gradients multiply to −1.
- Line through two points in the form $ay = bx + c$ with integers (Jun 25): find the gradient, substitute, clear fractions.
- Tangent to a circle centre O at P (Nov 23): gradient of OP = −5/4 at P(8, −10), so the tangent gradient is 4/5; intercepts Q(20.5, 0), R(0, −16.4); RQ = 26.3.
- Plotting curves: complete the table accurately, plot within half a square, join with a smooth curve.
- Transformations: $f(x + a)$ moves left a; $f(x) + b$ moves up b; $f(ax)$ divides x-coordinates by a; $-f(x)$ reflects in the x-axis. Minimum (8, −12) → $f(x) + 3$: (8, −9); $f(2x)$: (4, −12) (Jun 25).
- Trig graphs: $y = 2\sin(x + 30)°$ stretches by 2 vertically and moves 30° left (Jun 19).

9 CALCULUS

DIFFERENTIATION

$d/dx(ax^n) = nax^{n-1}$. Gradient at a point: substitute x into dy/dx. Turning points: dy/dx = 0; nature from d^2y/dx^2 or the shape.

Jun 24: $y = x^3 - 40x + 1$, gradient $8 \Rightarrow 3x^2 - 40 = 8 \Rightarrow x = \pm 4 \Rightarrow (4, -95)$ and $(-4, 97)$.

Jun 26 (maximum volume): $V = x \cdot 3x \cdot (13.5 - 2x) = 40.5x^2 - 6x^3$; $dV/dx = 81x - 18x^2 = 0 \Rightarrow x = 4.5 \Rightarrow V = 273.375\ cm^3$.

Kinematics: v = ds/dt, a = dv/dt; at rest when v = 0.

Tangent at a minimum (Nov 24): the gradient is 0, so the tangent is y = (minimum y-value).

10 GEOMETRY AND TRIGONOMETRY

TRIGONOMETRY TOOLKIT (SINE AND COSINE RULES AND ½AB SIN C ARE ON THE FORMULAE SHEET)

Right-angled: SOH CAH TOA, Pythagoras. Non-right: sine rule for a pair of opposite sides and angles; cosine rule for two sides and the included angle, or three sides.

Jun 25: $BD^2 = 9.4^2 + 12.8^2 - 2(9.4)(12.8)\cos 72° \Rightarrow BD = 13.3$; then sine rule BC = 13.3 sin 39°/sin 54° = 10.4.

3D (Jun 26, Nov 25): find the right-angled triangle containing the angle between the line and the plane; calculate its sides in the base first.

Bearings: measured clockwise from north; use parallel north lines (co-interior angles add to 180°).

CIRCLE THEOREMS (STATE EACH REASON)

Angle at the centre is twice the angle at the circumference · angle in a semicircle is 90° · angles in the same segment are equal · opposite angles of a cyclic quadrilateral add to 180° · tangent ⊥ radius · tangents from a point are equal · alternate segment theorem.

- Polygons: exterior angle = 360/n; interior + exterior = 180°. **Jun 25:** exterior angle $18° \Rightarrow n = 20$.
- Isosceles trapezium (Jun 25): height 12 sin 60°, base pieces 12 cos 60° = 6, top 47 − 12 = 35, area = 426 cm^2.

11 MENSURATION AND SIMILARITY

FORMULAE

Arc length = $(\theta/360) \times 2\pi r$; sector area = $(\theta/360) \times \pi r^2$; segment = sector − $\frac{1}{2}r^2 \sin\theta$. **Jun 25:** sector with 140°, $r = 16 \Rightarrow 313\ cm^2$.

On the sheet: cone, sphere, cylinder, prism volumes and curved surfaces. Hemisphere = half a sphere (add πr^2 for the flat face). Frustum = large cone/pyramid − small one.

SIMILAR SHAPES

Length scale factor k ⇒ area × k^2, volume × k^3. **Jan 22:** heights 10 and 15 ⇒ volumes 8 : 27; difference 19 parts = 1197 ⇒ 1 part = 63 ⇒ volume of A = 504 cm^3.

From areas to volumes (Nov 20): take the square root of the area ratio, then cube.

12 VECTORS

VECTOR METHOD (4–5 MARKS)

Write the path to the unknown point two ways, introduce a scalar (λ or n), equate the coefficients of a and of b, and solve.

Jun 25: OA = 12a, OB = 8b, P is the midpoint of OA, AB : BR = 1 : 2. AB = −12a + 8b; PR = 6a + 3AB = −30a + 24b; PQ = −6a + nb = λ(−30a + 24b) ⇒ λ = 1/5, n = 4.8.

- Parallel vectors are scalar multiples; collinear points share a vector direction and a point.
- Regular hexagons (Nov 21): opposite sides equal vectors; an enlargement scale factor 2 doubles every vector.

13 STATISTICS

- Mean from grouped data: Σ(midpoint × frequency) ÷ Σf. **Jun 25:** 920 ÷ 100 = 9.2 km. Modal class: the interval, not a frequency.
- Cumulative frequency: plot at upper class boundaries; median at n/2; IQR = UQ − LQ at 3n/4 and n/4; draw lines on the graph.
- Histograms: frequency density = frequency ÷ class width. **Jun 25:** people over 40 minutes = 22 out of 126 = 11/63. Partial bars: take the fraction of the width.
- Combined means (Jan 22, Jun 21): total = mean × number; new value = new total − old total.
- IQR from a list (Nov 25): LQ is the (n + 1)/4th value, UQ the 3(n + 1)/4th.

14 PROBABILITY AND SETS

WITHOUT REPLACEMENT (3 MARKS) · JUN 25

20 counters: 9 red, 7 yellow, 4 green; take 3. P(exactly two the same) = 3 × [(9/20)(8/19)(11/18) + (7/20)(6/19)(13/18) + (4/20)(3/19)(16/18)] = 51/76. The × 3 counts the orders.

"At least one": 1 − P(none). Unknown x (Jun 26): form the probability in x, set it equal to the given value, solve the quadratic.

- Trees: multiply along branches, add between outcomes. Conditional: the second-stage probabilities change.
- Expected frequency = probability × number of trials. Probabilities sum to 1 (find x in a table).
- Sets: A ∩ B both; A ∪ B either; A′ not A; n(A) number of elements; C ⊂ A subset. Fill Venn diagrams from the intersection outwards.

15 FINAL A* CHECKLIST

1. Show every stage: many questions are "working required".
2. Do not change a correct answer by simplifying wrongly afterwards.
3. Percentages: use multipliers; reverse percentages divide.
4. Bounds: choose the correct bound for each part of the formula; justify a "suitable degree of accuracy".
5. Surds and fractions: show conjugate multiplication and the simplified final form.
6. Simultaneous equations and quadratics: show substitution, the three-term quadratic and its solution; pair values.
7. Vectors: two routes, equate coefficients of a and b.
8. Calculus: dy/dx, set to the given gradient or 0, solve, substitute back.
9. Circle theorems: give each reason in words. Leave 10 minutes to check accuracy, units and answer lines.